Extraneural Glioblastoma Multiforme Vertebral Metastasis.

PMID: 26704201 | Indexed in PubMed

Glioblastoma multiforme (GBM) is the most common malignant central nervous system tumor; however, extraneural metastasis is uncommon. Of those that metastasize extraneurally, metastases to the vertebral bodies represent a significant proportion. We present a review of 28 cases from the published literature of GBM metastasis to the vertebra. The mean age at presentation was 38.4 years with an average overall survival of 26 months. Patients were either asymptomatic with metastasis discovered at autopsy or presented with varying degrees of pain, weakness of the extremities, or other neurologic deficits. Of the cases that included the time to spinal metastasis, the average time was 26.4 months with a reported survival of 10 months after diagnosis of vertebral metastasis. A significant number of patients had no treatments for their spinal metastasis, although the intracranial lesions were treated extensively with surgery and/or adjuvant therapy. With increasing incremental gains in the survival of patients with GBM, clinicians will encounter patients with extracranial metastasis. As such, this review presents timely information concerning the presentation and outcomes of patients with vertebral metastasis.

Immune phenotypes predict survival in patients with glioblastoma multiforme.

PMID: 27585656 | Indexed in PubMed

Glioblastoma multiforme (GBM), a common primary malignant brain tumor, rarely disseminates beyond the central nervous system and has a very bad prognosis. The current study aimed at the analysis of immunological control in individual patients with GBM. Immune phenotypes and plasma biomarkers of GBM patients were determined at the time of diagnosis using flow cytometry and ELISA, respectively. Using descriptive statistics, we found that immune anomalies were distinct in individual patients. Defined marker profiles proved highly relevant for survival. A remarkable relation between activated NK cells and improved survival in GBM patients was in contrast to increased CD39 and IL-10 in patients with a detrimental course and very short survival. Recursive partitioning analysis (RPA) and Cox proportional hazards models substantiated the relevance of absolute numbers of CD8 cells and low numbers of CD39 cells for better survival. Defined alterations of the immune system may guide the course of disease in patients with GBM and may be prognostically valuable for longitudinal studies or can be applied for immune intervention.

Extra-CNS metastasis of glioblastoma multiforme to cervical lymph nodes and parotid gland: A case report.

PMID: 32983474 | Indexed in PubMed

Extra CNS metastasis of glioblastoma multiforme is extremely rare. We report a case of a 53-year-old Caucasian male who, after undergoing surgical resection and nine months adjuvant therapy, had a recurrence of the cancer with an infiltration expanding outside the cranium to the left maxilla, mandible and parotid gland.

A case of epithelioid glioblastoma with lung metastases in a young Cane Corso dog.

PMID: 39442362 | Indexed in PubMed

Astrocytomas are relatively common primary brain tumours of humans and companion animals. In dogs, they represent approximately 17-28% of primary central nervous system tumours. However, extracranial metastasis is extremely rare. This case report describes a grade IV astrocytoma (glioblastoma) in the cerebrum of a young Cane Corso dog with pulmonary metastases. The diagnosis was obtained via histopathological morphology and immunophenotyping, which showed strong positivity for glial fibrillary acidic protein, vimentin and connexin-43. The glioblastoma in this Cane Corso had epithelioid morphology with histological features of malignancy including high mitotic count, microvascular proliferation, serpentine necrosis and subventricular zone involvement. Epithelioid glioblastoma is a rare subtype that has only relatively recently been formally acknowledged in human medicine and it can also pose a diagnostic challenge in veterinary medicine.

Metastatic glioblastoma to the lungs: a case report and literature review.

PMID: 38864820 | Indexed in PubMed

Glioblastoma is the most common malignant primary brain tumor. Despite its infiltrative nature, extra-cranial glioblastoma metastases are rare. We present a case of a 63-year-old woman with metastatic glioblastoma in the lungs. Sarcomatous histology, a reported risk factor for disseminated disease, was found. Genomic alterations of TP53 mutation, TERT mutation, PTEN mutation, and +7/-10 were also uncovered. Early evidence suggests these molecular aberrations are common in metastatic glioblastoma. Treatment with third-line lenvatinib resulted in a mixed response. This case contributes to the growing body of evidence for the role of genomic alterations in predictive risk in metastatic glioblastoma. There remains an unmet need for treatment of metastatic glioblastoma.

An Extracranial Metastasis of Glioblastoma Mimicking Mucoepidermoid Carcinoma.

PMID: 29852299 | Indexed in PubMed

Glioblastoma (GBM) is the most common and aggressive primary malignant tumor of the brain and central nervous system. Extracranial metastases of GBM are rare, with few case reports published to date. The tumor cells of GBM show strong immunopositivity for glial fibrillary acid protein. A 47-year-old man without comorbidities presented with a 1-year history of an augmenting right parotid lump. A right total parotidectomy with selective neck dissection was performed. The hematoxylin-eosin-stained slice of a parotid lymph node collected intraoperatively revealed destruction of normal lymph node structure by medium-sized pleomorphic cells scattered in groups; their cytoplasm was lightly stained and pale. There were abundant myxoid stroma in the interstitial tissue. This characteristic mimicked mucoepidermoid carcinoma. An immunohistochemistry test demonstrated that the tumor cells were positive for glial fibrillary acid protein. A diagnosis of extracranial metastasis of GBM was made after confirmation with postoperative pathologic examination and the review of the intracranial resection specimen. We believe that this is the first reported case of extracranial metastasis of GBM resembling mucoepidermoid carcinoma in the microscope features. Pathologists and clinicians should be alert to this rare lesion and consider this differential diagnosis after excluding other common parotid lesions.

Case report of pulmonary metastasis in a male Wistar rat glioblastoma model.

PMID: 33627949 | Indexed in PubMed

Glioblastoma (GBM) is a highly aggressive central nervous system cancer. Its extracranial metastases have rarely been reported in the past few decades. Moreover, the pathogenesis of extracranial GBM metastases remains unclear. Here, we report a case of pulmonary metastasis in a male Wistar rat of C6 GBM model. This reported Wistar male rat was one of the experimental control group without any other intervention except for C6 GBM cells orthotopic implantation. On postoperative day 15, the animal which was reported in this study showed highly cellular, pleomorphic, tumor with nuclear atypia in the brain (Ki67, approximately 65.7%) and lungs (Ki67, 49.5%). Tumor cells in the lung showed immunoreactivity for glial fibrillary acidic protein. Inflammatory CD68+ cell infiltration, weakly positive E-cadherin, and strongly positive staining for vimentin were observed both in tumors in the brain and lungs. Based on further morphological analysis, we speculate that the potential metastatic route into the lung might be hematogenous metastasis.

Extracranial metastasis of gliobastoma: Three illustrative cases and current review of the molecular pathology and management strategies.

PMID: 26137254 | Indexed in PubMed

Glioblastoma (GBM) is the most common and the most malignant primary brain tumor in adults, accounting for ~12-15% of all intracranial neoplasms. Despite advances in surgical, medical and radiation therapies, the mortality of GBM remains high, with a median survival ranging between 40 and 70 weeks. Similar to other primary brain tumors, the extracranial metastasis of GBM is extremely rare, occurring in <2% of patients. To demonstrate the clinical characteristics of this rare tumor, we herein present three cases of extracranial GBM metastasis: One to the lungs, which represents the longest reported survival of lung metastases from GBM to date; the second to the soft tissue of the posterior neck; and the third to the lumbar intradural space. Unlike tumors elsewhere, there are unique barriers in the brain that prevent the hematogenous and lymphatic spread of intracranial tumors, such as the dura mater and the thickened basement membrane of the blood vessels. In addition, central nervous system tumor cells lack extracellular matrix proteins required to invade surrounding connective tissue, a prerequisite for tumor dissemination. In this study, we aimed to investigate the different possible mechanisms underlying the extracranial metastasis of GBM and determine the biomolecular and genetic characteristics differentiating GBMs that metastasize from those that do not. We also reviewed the role of systemic chemotherapy and bevacizumab in the treatment of disseminated GBMs. Early identification and differentiation of these tumors may enable patients to benefit from surgical resection, radiation and combination chemotherapy prior to developing other comorbidities from metastatic disease, which may translate into prolonged survival with an acceptable quality of life.

GBM skin metastasis: a case report and review of the literature.

PMID: 28718312 | Indexed in PubMed

Glioblastoma (GBM) is the most common type of malignant tumor found in the brain, and acts very aggressively by quickly and diffusely infiltrating the surrounding brain parenchyma. Despite its aggressive nature, GBM is rarely found to spread extracranially and develop distant metastases. The most common sites of these rare metastases are the lungs, pleura and cervical lymph nodes. There are also a few case reports of skin metastasis. We present the clinical, imaging and pathologic features of a case of a GBM with metastasis to the soft tissue scar and skin near the original craniotomy site. In addition, we discuss the details of this case in the context of the previously reported literature.

Extracranial metastases in secondary glioblastoma multiforme: a case report.

PMID: 33087049 | Indexed in PubMed

Glioblastoma (GBM) is known for its devastating intracranial infiltration and its unfavorable prognosis, while extracranial involvement is a very rare event, more commonly attributed to IDH wild-type (primary) GBM evolution. We present a case of a young woman with a World Health Organization (WHO) grade II Astrocytoma evolved to WHO grade IV IDH mutant glioblastoma, with subsequent development of lymphatic and bone metastases, despite the favorable biomolecular pattern and the stability of the primary brain lesion. Our case highlights that grade II Astrocytoma may evolve to a GBM and rarely lead to a secondary metastatic diffusion, which can progress quite rapidly; any symptoms referable to a possible systemic involvement should be carefully investigated.